

Spatial aggregation alters population-weighted estimates of InSAR observation support

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Abstract

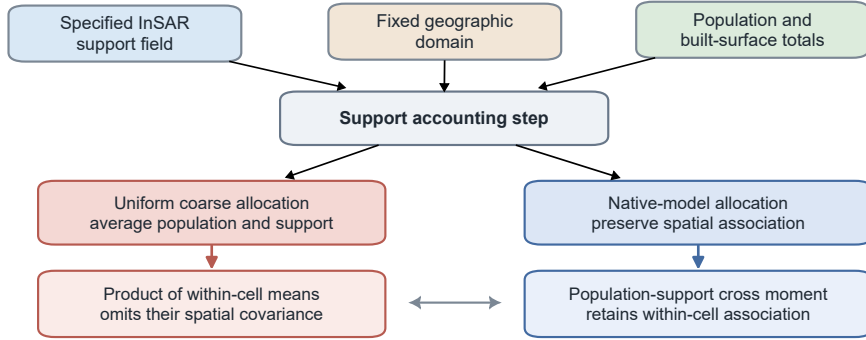
Population-weighted assessments of interferometric synthetic aperture radar (InSAR) observation support depend on how population is allocated within reporting units. Spatial aggregation can change these assessments even when the observations are unchanged. We isolate this effect by comparing allocation rules under a fixed geographic domain, support field and population total. In the Chao Phraya footprint, uniform allocation within 1-km cells increases the mean support deficit by 57.7–57.9% relative to native population-model allocation, equivalent to 7.75–7.78 percentage points of the domain total. Within-cell covariance between population and support explains this difference, which persists after excluding boundary cells. Crossed population-model and interferogram-sampling comparisons preserve its direction but change its magnitude. Shifts as a share of total population are smaller for the final processing mask and in the California cases. Controlled missingness shows that small errors where estimates remain available can coexist with substantial unestimated population allocation. The primary deficit averages unsupported model allocation over area and interferogram pairs; it does not identify subsidence exposure. Population-weighted reporting should therefore preserve the spatial association between population and observation support and report sensitivity to allocation and sampling.

Keywords: InSAR, population allocation, observation support, coherence, spatial scale, reproducibility

047 1 Introduction

048
049 Satellite radar missions such as Sentinel-1 provide repeated observations for
050 deformation mapping [1]. Classical reviews describe its interferometric basis and
051 geophysical applications [2–5]. Persistent-scatterer, small-baseline and later multi-
052 temporal methods extend monitoring across point-like and distributed scatterers
053 [6–9]. Public processing chains and regional products have broadened access to
054 urban and deltaic subsidence measurements [10–12]. For downstream screening,
055 deformation layers must be translated into population summaries without losing the
056 support limitations recorded in the product.
057 This translation requires spatial accounting because InSAR does not observe all sur-
058 faces equally. Temporal decorrelation and surface change can reduce measurement
059 validity [13]; vegetation, water, construction, geometric effects, coherence thresh-
060 olds and quality masks further structure product support [14]. Deformation can also
061 vary within cities and engineered landscapes [15, 16]. Intersecting population, built-
062 up land or infrastructure with valid radar support defines a support-conditioned
063 domain, appropriate for a target explicitly conditional on observed locations. A
064 broader-domain summary must additionally report allocation outside that support.
065 Missing support alone does not establish deformation at omitted locations.
066 Coastal and urban studies link subsidence to exposure and flood risk [17, 18],
067 while infrastructure and megacity analyses examine the built environment [19, 20].
068 Global and regional reviews document its geographic extent and the role of satel-
069 lite interferometry [21–24]. Nationwide inventories, building-level diagnostics and
070 coastal population-weighted assessments address distinct physical and spatial tar-
071 gets [25–27]. Their population summaries depend on both the observation domain
072 and population allocation; screening workflows must state what remains unresolved
073 outside observed support.
074 Published assessments also combine quality screening with geodetic comparisons.
075 The Mexico-wide assessment applied temporal-coherence screening and compared
076 InSAR estimates with GPS observations [28], and Ohenhen et al. evaluated rates
077 against 154 GNSS stations across 28 US cities [29]. Published Bangkok studies
078 provide further leveling and CORS comparisons, with their spatial, temporal and
079 station-specific qualifications detailed in Supplementary Section S3 [30–32]. Mur-
080 ray et al. also noted that motion between buildings may remain undetected and
081 require additional observations [33]. These practices and stated limitations pro-
082 vide an essential basis for interpreting displacement measurements and their spatial
083 support.
084 Population allocation introduces a further step when displacement products are
085 combined with demographic data. Blackwell et al. estimated community exposure
086 assuming uniform population within each community, while Ohenhen et al. assigned
087 census-block population to subsidence classes using each block’s median InSAR-
088 pixel rate [29, 34]. Both studies also used GNSS comparisons. Those comparisons
089 assess displacement agreement; the allocation rules determine how measurements
090 are translated into population summaries. We examine this latter step under a spec-
091 ified observation-support field, holding the domain, support and population total
092

fixed. These published choices motivate the comparison without implying that either study mishandled missing observations. Integrating these datasets requires distinguishing the radar support, the physical deformation screen and the population or built-surface layer. Allocation outside the analysis domain differs from allocation whose deformation status is unresolved. Reproducibility requires recording mask application and denominator changes alongside dataset provenance. Downstream users may combine a released quality mask with a different population grid or reporting geography. Spatial support is part of the data model [35], and aggregation can alter the inferred quantity as recognized by the modifiable areal unit problem [36]. Our matched-input comparison tests whether within-unit population-support association is preserved or replaced by uniform allocation. Areal interpolation and dasymetric mapping provide established approaches to allocating population within coarse units [37, 38]. A finer grid preserves more of the selected model’s spatial structure, without establishing household accuracy. Replacing the average population-support product with the product of separate averages omits within-unit covariance. This established identity explains a possible aggregation effect; it does not determine its empirical size. The questions are whether the effect persists away from boundaries, across population models and support definitions, and when missingness leaves some reporting units without an estimate. We treat public-product observability explicitly, keeping the geographic domain, radar support and population or built-surface allocation separate until accounting. Continuous mean area-pair support replaces the majority threshold as the primary definition. The support deficit measures model allocation unsupported on average across the specified observations; it neither identifies unique unobserved residents nor confirms subsidence without measurements. The earlier binary formulation remains a historical comparator in Section 2.3 and Supplementary Section S2. The audit addresses population-weighted accounting conditional on product support. It does not assess whether displacement measurements are accurate where they exist. Figure 1 summarizes the comparison of uniform coarse allocation and native-model allocation under identical inputs, with threshold, allocation and partition sensitivity reported explicitly [39].



139 We evaluate population-weighted support accounting under matched inputs. Scale,
140 origin and complete-interior comparisons separate allocation effects from bound-
141 ary handling. Crossed population-model and pair-selection tests and a same-grid
142 comparison of input support and final quality test dependence on data and sup-
143 port definitions. Controlled missingness evaluates conditional error and unestimated
144 allocation; regional cases test transfer. Together, these comparisons identify when
145 uniform allocation is adequate and when population–support association mat-
146 ters. The reproducibility package records inputs, denominators and executable
147 comparisons [40].
148 We report small effects, sign reversals and unestimated allocation alongside the pri-
149 mary result. Physical interpretation requires geodetic and geotechnical evidence
150 matched to the locations and quantities being assessed.

151

152 2 Materials and methods

153

154 2.1 Task formulation and notation

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156 The workflow combines a declared geographic domain, reporting grid, support field
157 and population or built-surface layer. It returns supported and unsupported allo-
158 cation and their sensitivity to support and allocation choices. This accounting is
159 downstream of displacement retrieval; the separate processing experiment below
160 evaluates one specified LiCSAR/LiCSBAS chain.

161 We distinguish the geographic domain D (area of interest, AOI), support $s(x) \in$
162 $[0, 1]$, and nonnegative exposure density $\rho(x)$. The primary Chao Phraya domain
163 contains the 1,807,700 GHSL 3-arc-second pixels whose centers lie in the frozen
164 finite footprint of 18,077 archived VLM cells. It is held fixed without velocity-
165 threshold selection. Because the boundary inherits the finite source-product
166 footprint, it does not represent complete coverage of the geographical delta.
167 The archived VLM GeoTIFF matches the Chao Phraya file in the global-delta
168 dataset accompanying Ohenhen et al. [41, 42]. That study used WabInSAR. Our
169 LiCSAR quality sample is a different processing chain and cannot be interpreted as
170 the validity mask of the WabInSAR product. Moreover, the public GeoTIFF lacks
171 an explicit stored-unit tag establishing the archived factor-of-ten conversion. We
172 therefore retain the inherited strong-deformation selection only as a clearly labeled
173 supplementary sensitivity. Neither that conversion nor a cross-product quality proxy
174 determines the primary geographic population denominator.

175 For pair j , $M_j(x) = 1$ where unwrapped phase is finite, raster-valid and nonzero
176 and, for the joint screens, byte coherence divided by 255 is at least q . The zero rule
177 is product-specific [43]. We evaluate $q = 0.2, 0.3, 0.4$ and unwrapped validity alone.
178 Integration uses each raster’s actual GDAL transform and intersects masks before
179 averaging; separately averaged masks would not yield their spatial intersection.
180 For GHSL pixel p with area A_p , the reported support is

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$$s_p = \frac{1}{NA_p} \sum_{j=1}^N \int_p M_j(x) \, dA. \quad (1)$$

Thus s_p is mean area-pair support at the exposure grid. It is distinct from the area whose individual locations pass a fixed number of pairs, from a binary majority classification at a cell center, and from final inversion validity. We assume each GHSL pixel's population or built-surface total is uniform within that pixel when integrating its intersections. Spherical rectangular areas use longitude and sine-latitude boundaries.

With nonnegative allocated exposure E_p , supported allocation is $S_E = \sum_p E_p s_p$, mean support deficit is $U_E = \sum_p E_p (1 - s_p)$, and total allocation is $T_E = \sum_p E_p$, so $S_E + U_E = T_E$. A pixel assigned 100 people and mean support 0.8 contributes 20 population-allocation units to U_E . These are averages over area and pairs, not 20 identifiable residents. Neither population-valued nor built-surface-valued deficits establish deformation, vulnerability or damage.

The weighted support distribution is

$$F_E(t) = \frac{\sum_p E_p \mathbf{1}(s_p < t)}{T_E}, \quad \int_0^1 F_E(t) dt = \frac{U_E}{T_E}. \quad (2)$$

We report the full support distribution. Its integral is checked using sorted support values, not trapezoidal integration of the plotted 101-point curve.

2.2 Input schema and reproducibility contract

Each run requires stable cell identifiers, geometries and areas; a geographic domain declared before accounting; documented observation support; and population or built-surface quantities on a known spatial support. Any deformation selection additionally requires established units and a stated role. Raster masks, measurement polygons and point observations need distinct support definitions. A point is not repeatedly assigned the full population of its containing cell; an areal rule must first be specified and checked for overlap.

A clipped raster without its removed-support record cannot identify the omitted domain. Likewise, population without geographic support cannot be allocated, and infrastructure counts without asset metadata do not establish engineering risk. These input limits prevent a numerical total from being interpreted beyond its target.

Manifests record source versions, transforms, nodata conventions, established units, dates and checksums. Raster validity, coherence, final quality masks and polygon coverage remain separate fields. Intermediate outputs retain the domain, allocation rule, scale, origin and denominators, with per-pair, per-block or per-station records as applicable (Supplementary Sections S1, S4–S5 and S7). Transfers preserve extensive totals on the common domain.

2.3 Workflow architecture

The binary cell formulation separates deformation-screen membership from whether sampled support passes a majority cutoff (Supplementary Fig. S1). This partition defines a coarse-cell selection: a majority threshold does not identify permanently

231 unobserved locations, and a selected cell does not locate residents over subsiding
 232 ground. We retain the original classification as a historical comparator. Continuous
 233 support on the fixed geographic domain defines the primary analysis.
 234 Module 1 registers inputs to the common domain, retaining included area in partial
 235 boundary cells. Module 2 intersects validity and coherence masks before integrating
 236 support at native population resolution. Module 3 fixes geographic and population
 237 denominators independently of allocation; the historical deformation-selected par-
 238 tition remains in Supplementary Fig. S1 and Supplementary Section S2. Module
 239 4 computes supported and unsupported allocation under uniform-area and native-
 240 model rules with identical totals. Module 5 tests sensitivity to support definition,
 241 sampling, population model and reporting units. Recorded input selections define
 242 each run; thresholds are not chosen to maximize the deficit.
 243 The center, area-weighted and ancillary-weighted rules represent distinct choices
 244 within established areal-interpolation practice [37, 38]. Supplementary Section S6
 245 states their information requirements and which comparisons are actually per-
 246 formed. Native GHSL allocation is the common model reference for these tests;
 247 comparisons with deformation imputation or multi-sensor retrieval would require a
 248 different target and additional observations.
 249 Extensive first and cross moments are conservatively transferred to a 50-m UTM
 250 integration grid (zone 47N) and repartitioned into 250-, 500-, 1,000- and 2,000-m
 251 cells. Each size uses four origins, with zero and half-cell shifts. Partial cells retain
 252 only included area; population is not extrapolated outside the domain. A 25-m
 253 recomputation checks integration sensitivity (Supplementary Sections S7 and S18).
 254 Within cell c , let $P_c = \int_c \rho \, dA$, $H_c = \int_c \rho s \, dA$, $A_c = \int_c dA$ and $Q_c = \int_c s \, dA$. Native-
 255 model reference deficit is $P_c - H_c$, while uniform allocation gives $P_c(1 - Q_c/A_c)$.
 256 Their difference is

$$257 \quad U_{\text{uniform}} - U_{\text{native}} = \sum_c \left(H_c - \frac{P_c Q_c}{A_c} \right) = \sum_c A_c \text{Cov}_{A,c}(\rho, s). \quad (3)$$

261 Preserving H_c retains the population–support association. Center sampling instead
 262 uses the fine pixel at each coarse center, with zero support outside the domain. We
 263 report signed and absolute discrepancies relative to the native-model deficit and
 264 fixed AOI total. A complete-interior comparison excludes boundary-crossing cells,
 265 with separate subset denominators and containment checked against the trans-
 266 formed native-pixel boundary (Supplementary Section S7). The native reference is a
 267 population model, not household truth.

269 2.4 Study design and evidence roles

270 Chao Phraya supplies the primary allocation comparison, a separate Bangkok rect-
 271 angle the connected processing experiment, and Fresno a published observation
 272 footprint. Los Angeles and Davis–Sacramento test additional product and regional
 273 transfer. These designs have separate support definitions and validation targets.
 274 In a fixed Bangkok rectangle (100.35–100.65° E, 13.65–14.05° N), 437 pairs from
 275 2019–2020 with baselines at most 96 days connect 104 acquisition dates. LiCSBAS2
 276

steps 11–16 use the aligned native inputs and fixed defaults without automatic	277
threshold relaxation [10, 44]. Before pixel-quality inspection, 44 edges are randomly	278
held out while preserving date connectivity; 393 training pairs undergo the same	279
processing. Training time-series differences predict relative LOS displacement at	280
the held-out edges, with training-linear and zero-displacement baselines on common	281
evaluable observations. Referencing, exclusions and exact parameters are recorded in	282
Supplementary S17.	283
Support strata and quality masks use training inputs only. Held-out dates remain	284
represented in training, so shared dates and atmospheric phase restrict interpre-	285
tation to internal edge reconstruction, not independent velocity accuracy or fore-	286
casting. Separately, legacy 28-pair and matched-stack support are compared with	287
full-stack quality, with GHSL counts transferred by source-area fractions. Coherence	288
contributes to mask construction, and the legacy period differs. This evaluates the	289
specified LiCSAR/LiCSBAS chain, not the WabInSAR raster; matched independent	290
Bangkok GNSS validation remains unavailable.	291
A fixed Fresno rectangle (119.9–119.7° W, 36.6–36.85° N) tests transfer to the DWR	292
2026Q1 observation-cell footprint [45, 46]. We clip and union all 38,738 intersecting	293
measurement polygons, then allocate GHSL population once to their nonduplicated	294
intersections. Scale/origin comparisons use UTM zone 11N. The polygon layer sup-	295
plies coverage, not per-cell velocity or uncertainty; its 2015–2026 observation record	296
also differs from the 2020 population epoch. Artificial masking is restricted to the	297
published footprint.	298
	299
<i>Population agreement and external transfer designs.</i>	300
Fresno’s irregular 2020 Census blocks use GHSL allocation normalized to P1 totals	301
and known DWR rates for 1 October 2020–1 October 2021. Random masks and one	302
or four contiguous holes remove 10, 30 or 50% of known-rate cells, with 30 repeti-	303
tions at -3 , -5 and -10 mm yr $^{-1}$. Native/area prevalence and unit-median rules	304
share masks; undefined allocation, common-computable errors and whole-domain	305
bounds are reported separately (S15).	306
Five-band spatial holdouts with 0-, 1- and 3-km buffers test training-only total	307
recalibration against ACS, not original population-model training. Shared-date	308
omissions assess dependence in the 224-pair pool without velocity re-inversion (S20).	309
Eighteen training-only quality rules evaluate coverage and residuals on 44 excluded	310
edges; held-out errors do not select thresholds (S21).	311
Fixed Los Angeles and Davis–Sacramento rectangles test three cutoffs, four scales	312
and four origins using archived ARIA products [47, 48] (S29). Fresno municipal land	313
use supplies strict/broader residential eligibility with unmapped areas retained [49]	314
(S27). Archived NGL GNSS LOS comparisons retain both reference frames and	315
three sampling radii, excluding provider and location-selected local controls from	316
all validation metrics. Temporal-model and source-version limits are retained; this	317
external-product check neither refits daily GNSS nor validates Bangkok (S30).	318
	319
	320
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323 2.5 Exposure layers and allocation

324 GHSL supplies consistent public population and built-surface layers across the
325 study domains. Comparing uniform-area and native allocation isolates a spatial
326 assumption under identical inputs; neither represents household-level validation. The
327 resulting quantities are model allocations rather than official demographic or loss
328 estimates.

329 We compare GHSL R2023A epoch-2020 population at 3 arcsec with WorldPop’s
330 Thailand 2020 unconstrained UN-adjusted population-count product at 3 arcsec
331 [50, 51]. Source counts are transferred by spherical source-area fractions, not by
332 interpolation of density values as if they were counts. Native totals, jointly valid
333 coverage and common-total normalized results are all reported. GHSL 2015/2020
334 epoch comparisons hold resolution at 30 arcsec. Neither population product is local
335 census truth, and epoch differences combine demographic and model effects.
336

337 *Crossed aggregation sensitivity.*

338
339 We cross GHSL and WorldPop 2020 with legacy 28-pair and expanded 224-pair
340 support at $q = 0.3$. All combinations use the same two-model common-valid
341 domain and GHSL total, with 1-km cells at four origins and conserved first and
342 cross moments. Differences are reported relative to both the native deficit and com-
343 mon population total. The 25-m check, all 16 cases and pair-selection details remain
344 in Supplementary Section S18; changing pair number also changes composition.
345 WorldCover 2021 v200 supplies 10-m land classes [52]. Their fractional areas within
346 GHSL pixels inherit the within-pixel uniform-allocation assumption. Tree-class allo-
347 cation does not identify residences, and land-cover, population and radar epochs
348 differ.
349

350 2.6 Robustness and bounded-evidence protocol

351 The legacy sample contains 28 pairs from 2016–2022. The expanded pool selects
352 eight pairs per year–quarter stratum (224 total) using dates and baseline classes
353 before outcome extraction. Nested ranks yield 28, 56, 112 and 224 pairs; 100 strat-
354 ified draws without replacement per size assess conditional sampling variation. A
355 pre-extraction rank amendment and both rank plans are archived. Year, quarter,
356 baseline and shared-date omissions test composition and dependence. These com-
357 parisons do not estimate independent-observation confidence bounds or uncertainty
358 outside the fixed pool (Supplementary Sections S16 and S20).

359 Controlled masking uses a separate fixed Bangkok rectangle (100.35–100.75° E,
360 13.55–13.95° N). Random, spatially smoothed and positive/negative log-density-
361 associated scores remove 10%, 30% or 50% of fine pixels, with a fixed seed and 100
362 repetitions. Uniform-area, center and built-area-weighted rules share masks and
363 coarse totals; built weights fall back to area where zero. Grids contain 5, 10 or 20
364 native pixels at two origins, not exact metric cells.

365 Errors use the known removed allocation U_{ref} . Percentile ranges describe introduced-
366 mask variation conditional on the fixed inputs, not population or InSAR accuracy.
367 Built weighting shares information with GHSL and is not independent validation.
368

We also construct three synthetic hazard fields: positive density association, negative association and an independent spatial field. Rates of -10 or 0 mm yr^{-1} define known threshold membership. For observed threshold allocation T_{obs} and allocation U in pixels missing under a binary final mask, the identification interval is $[T_{\text{obs}}, T_{\text{obs}} + U]$. Its endpoints assign all missing allocation outside or inside the class. Here U is binary-mask missing allocation, not the continuous mean area-pair deficit. The bound is conditional on the observed rates and population model; it is neither a physical prediction nor a confidence interval.

3 Results

3.1 Lead-case support accounting

Figure 2 compares native-model and uniform allocation under the same support field, geographic domain and allocated totals.

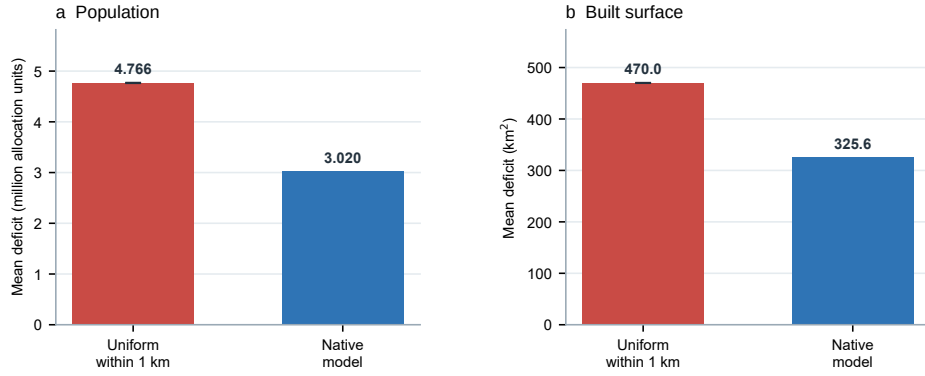
The fixed Chao Phraya footprint contains 22.501 million GHSL population units and 1253.6 km^2 of built surface. At $q = 0.3$, native allocation gives $U_E = 3.020$ million population units (13.42%) and 325.6 km^2 of built surface. Population is concentrated in relatively well-supported portions of the footprint (Supplementary Fig. S14). Consequently, area-, built-surface- and population-weighted support distributions differ (Fig. 3).

Table 1 Native-model support deficits.

Support rule	Population (million units)	Population (%)	Built surface (km^2)	Area (%)
Valid unwrapped	0.268	1.19	31.9	17.89
Joint, $q = 0.2$	1.665	7.40	197.5	57.44
Joint, $q = 0.3$	3.020	13.42	325.6	68.77
Joint, $q = 0.4$	4.891	21.74	464.6	76.83

Unwrapped validity alone and the coherence-screened quantities are substantially different (Table 1). Raising q from 0.2 to 0.4 changes the mean population deficit from 1.665 to 4.891 million. This range is a definition sensitivity, not a confidence interval. The weighted support curves retain information lost by applying one majority threshold to every coarse cell.

Supplementary Section S2 retains the original 3.77-million statistic and input lineage. Its center-majority selection and domain differ from the present mean deficit; the new value is not an estimate of the same quantity or a count of confirmed subsidence exposure.



3.2 Sensitivity to support definition, allocation and population models

With support, native population and AOI total held fixed, uniform allocation within 1-km cells yields 4.763–4.770 million mean population units in deficit, compared with the native-model reference of 3.020 million. This is 57.7–57.9% higher across the four origins. Aggregate allocation discrepancy grows across the examined 250–2,000-m scales (Supplementary Fig. S15). The center approximation also changes with scale and origin; its variability cannot be interpreted as a measurement uncertainty distribution.

The signed discrepancy follows the population–support covariance in Eq. 3. Positive aggregate covariance means uniform redistribution moves population away from better-supported locations, although individual cells can have either sign. Using 25-m instead of 50-m integration changes the 1-km estimate by at most 0.00024 percentage points of AOI population. A separate pixel-center diagnostic verifies the identity and its Cauchy–Schwarz bound (Supplementary Sections S8–S9).

The 1-km effect persists when all boundary-crossing cells are excluded. Across four origins, the complete interior cells contain 99.27–99.49% of domain population, and uniform allocation exceeds their own native-model deficit by 58.24–58.56%. Interior cells account for 98.55–98.87% of the full-domain signed uniform-allocation difference. Thus boundary treatment does not explain the primary allocation effect. Supplementary Section S7 separates complete, partial and outside-center cells and reports both methods under each origin.

On pixels jointly valid in the four population/epoch layers, the $q = 0.3$ deficit is 13.37% for native GHSL 2020 and 23.20% for WorldPop 2020. Joint coverage excludes 142.9 thousand GHSL population units from the full primary domain. Equalizing the two models to the same joint-domain total gives 2.989 versus 5.186 million mean deficit units, so total-population differences do not explain the discrepancy. At fixed 30-arc-second resolution, GHSL’s 2015 and 2020 shares are 20.86% and 20.28%. These epoch sensitivities are smaller than the native-model contrast here, without establishing which model is accurate (Supplementary Fig. S16).

Crossing population model and pair selection.

On the fixed GHSL–WorldPop common domain, both models are normalized to 22.358 million population units. At 1 km, uniform allocation exceeds the native-model deficit by 55.38–55.64% for GHSL with the legacy 28 pairs and 51.29–51.51% with the expanded 224 pairs. The corresponding WorldPop differences are 9.03–9.07% and 8.64–8.68%. All 16 model–sample–origin combinations have positive differences; normalized by the common total, these span 7.40–7.53 percentage points for GHSL and 2.09–2.13 for WorldPop. The direction therefore persists across these inputs, while its magnitude is strongly population-model dependent. The original approximately 58% difference is specific to its full-domain GHSL case. Changing 28 to 224 pairs changes selection and composition as well as sample size; the comparison is not an isolated sample-size effect. Supplementary Section S18 reports every origin and the numerical convergence check.

The Fresno population comparisons separate source totals from spatial allocation. Against 2020 Census blocks, total-normalized 1-km GHSL and WorldPop allocations have weighted absolute percentage error (WAPE; summed absolute unit discrepancies divided by the Census total) of 57–59%, below an area-only baseline (Supplementary Section S24). A 100-m comparison normalizes GHSL and the California population surface (CA-POP) to the same Census block counts; the outside-footprint allocation differs by 0.143 pp, with mixed tract-level directions (Supplementary Section S25). CA-POP uses the same Census counts as inputs, so this is an allocation sensitivity.

Against ACS 2017–2021 estimates for 319 complete Fresno block groups, Census P1, GHSL and WorldPop totals differ by -0.36% , $+10.57\%$ and $+6.03\%$, respectively. Census falls within the aggregate ACS 90% margin of error, without validating household locations. Most block groups are smaller than the 1-km source pixels, so source-cell uniformity remains material. Buffered spatial recalibration gives similar in-sample and held-out errors (Supplementary Sections S20, S26); these reporting-unit checks do not establish fine-scale accuracy.

Municipal residential eligibility leaves about one-third of model allocation unclassified. Its restricted common domain covers 89.13% of complete Census population, below the exploratory 90% reporting gate. Reduced deficits on that domain therefore support only a limited-domain sensitivity (Supplementary Section S27).

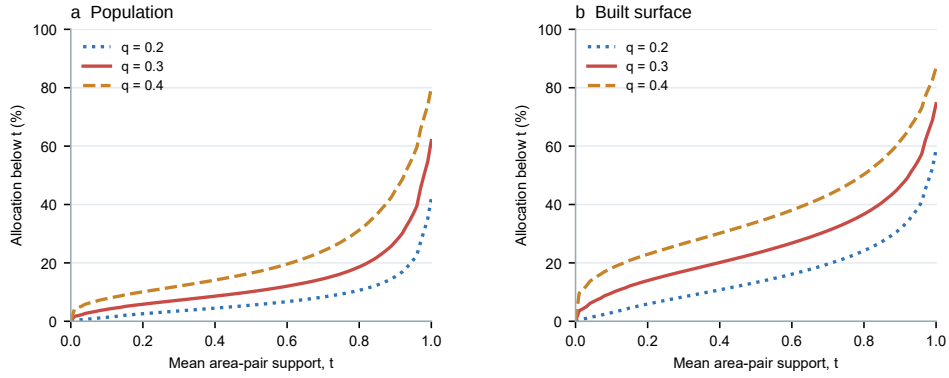
3.3 Sampling uncertainty and its scope

Figure 4 distinguishes conditional variation over draws from the fixed nested selections.

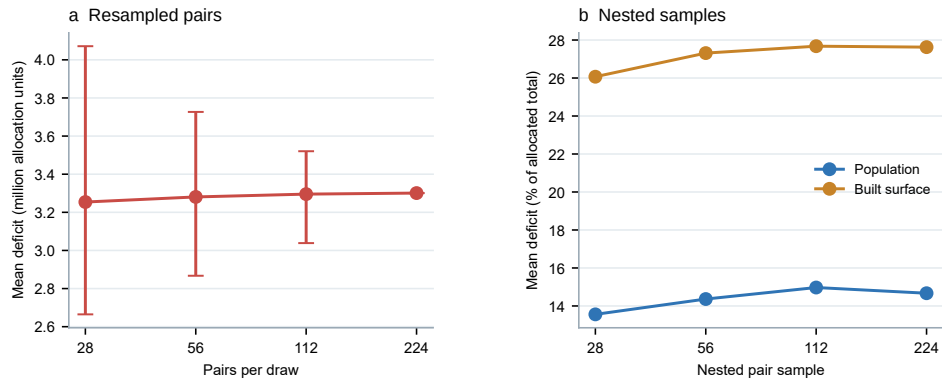
The legacy 28-pair date graph contains 54 dates, 26 components and no independent cycles. It cannot stand in for a complete time-series inversion network. The 224-pair sampling graph is also not fully connected: it has 225 dates and 29 components. In contrast, the 2016–2022 catalogue graph has 288 dates and one component.

Catalogue-level connectivity does not establish pixel-level validity.

The nested 28-, 56-, 112- and 224-pair means give population deficits of 3.051, 3.232, 3.369 and 3.301 million, respectively (Supplementary Fig. S4). The nested 28-pair subset is selected from the expanded pool and differs from the original 28 pairs used



in the primary 3.020-million calculation. Legacy leave-one-year-out results range from 2.820 to 3.181 million. Within the fixed 224-pair pool, the central 95% range of the 100 stratified 28-pair draws is 2.665–4.072 million. That conditional sampling range is wider than the differences between several individual nested means and cannot be converted into a confidence interval for final product quality. Longer-baseline groups have larger mean deficits in both samples. Differences between sample sizes also reflect changes in baseline and seasonal composition. Deleting each shared acquisition date changes the $q = 0.3$ equal-quarter population deficit by -0.386 to $+0.165$ pp. These are dependent support sensitivities, with no velocity re-inversion or confidence interval (Supplementary Section S20).



The checks are summarized by the alternative explanation they address (Table 2). Their different targets preclude treating them as a single validation score.

Table 2 Checks on the accounting result and their interpretation.

Check	Result or comparison	Interpretation
Support definition	Same-grid uniform-minus-native shift: input support 3.066–3.189 pp; final mask 0.108–0.117 pp	Input support is not final-product validity
Scale and grid origin	Complete-interior 1-km difference: 58.24–58.56% of native deficit	Boundary handling does not explain the primary effect
Population and pair selection	Positive shift in all 16 GHSL/WorldPop and 28/224-pair origin cases	Direction persists; magnitude depends on the population model
Controlled missingness	At 30% removal, irregular- block prevalence leaves 0.20% unestimated for random pixels versus 23.01–28.64% for holes	Conditional error and estimator availability must be reported together
Sampling dependence	Fixed-pool repetitions and shared-date omissions	Conditional sensitivity; not independent-observation confidence bounds
Processing and transfer	Excluded pair edges, DWR geometry and external California GNSS	Different evidence targets, summarized in Supplementary Table S29

3.4 Descriptive ranking stability

Spatial rankings are more stable than the totals: across 539 eligible fixed geographic blocks, nested 28-, 56- and 112-pair deficit-amount ranks correlate 0.9940, 0.9992 and 0.9994 with the 224-pair comparator. Top-decile overlaps nevertheless change (Supplementary Fig. S5). This is descriptive stability, not validation of a measurement-priority rule.

Top-decile Jaccard overlap for deficit amount is 0.8000, 0.8621 and 0.9286 for nested 28, 56 and 112 pairs against 224 pairs. Supplementary Fig. S6 reports amount and share rankings. High all-block correlation can coexist with changed short-list membership; monitoring priority would additionally require deformation, consequences and information value.

3.5 Failure-mode experiments and estimator availability

For random and spatial masks unrelated to density, aggregate mean bias is small across repetitions, while the center method has larger sample-to-sample variation. Density-associated masking changes the sign and size of the error (Supplementary Fig. S17). At 30% removal and 10-pixel cells with zero origin, uniform allocation underestimates the known removed population by 14.17% when high-density locations are preferentially removed and overestimates it by 120.35% when low-density locations are preferentially removed. Large relative errors occur when the removed

599 population is small. We also report errors normalized by the total test-domain
 600 population.
 601 Built-area weights reduce error in these GHSL-based scenarios but retain nonzero
 602 errors and shared-input dependence. The synthetic hazard experiments additionally
 603 show that support deficit alone does not identify threshold-exposed population: the
 604 unknown allocation can contain very different hazard shares under the three con-
 605 structed velocity fields. The conservative identification interval contains the known
 606 truth by construction; its width expresses missing information rather than recovered
 607 motion. Fresno masking outputs repeat the same protocol and are supplied in the
 608 Supplementary Information.
 609 Small conditional errors can coexist with substantial unestimated allocation. At
 610 30% removal, 10-pixel units and -5 mm yr^{-1} , prevalence leaves less than 0.0001%
 611 unestimated under random masks but 11.35–28.66% under structured masks. On
 612 the common-computable domain, prevalence has smaller local L1 error than median
 613 classification for three mechanisms; the median rule is better under low-spread
 614 removal. Whole-domain bound widths remain 24.12–34.90 pp (Supplementary
 615 Sections S12, S14).
 616 Actual irregular Census blocks show the same availability problem. At 30% removal,
 617 prevalence leaves 0.20% unestimated under random masking, versus 23.01% under
 618 one contiguous hole and 28.64% under four holes. The median rule leaves 9.41%,
 619 26.80% and 32.61%, respectively. Its whole-block classification also differs from the
 620 fine-allocation target before masking. Complete error and bound results are given in
 621 Supplementary Section S15.
 622 For the actual LiCSBAS final mask, unsupported allocation is 0.2944% of the fixed
 623 domain total. At LOS rate $\leq -5 \text{ mm yr}^{-1}$, observed class allocation is 0.2797% and
 624 its whole-domain share is bounded by $[0.2797, 0.5741]\%$, conditional on the observed
 625 rates and population model. The 0.2944-pp interval is narrow relative to the domain
 626 total but its upper endpoint exceeds twice the observed class share. Both end-
 627 points are attainable by changing only the unknown class assignments. Thus a small
 628 unsupported share can still matter for a rare threshold class; no motion is recovered
 629 in the gaps (Supplementary Section S13).

630

631 3.6 Regional and product transfer

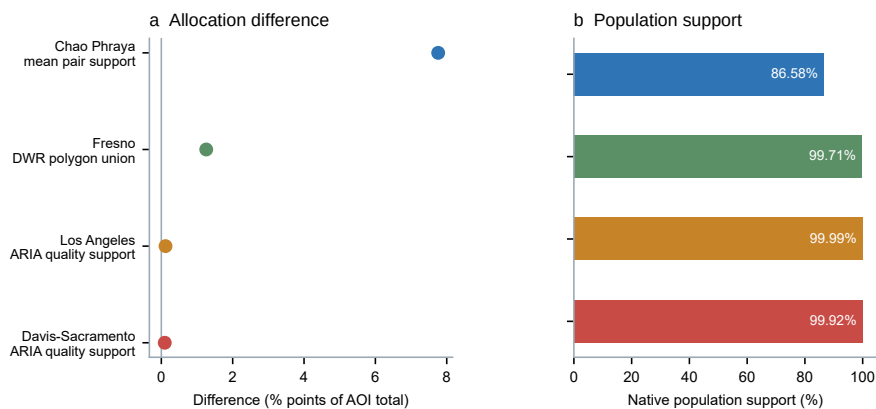
632 Figure 5 compares the allocation effect and population-weighted coverage under the
 633 declared regional support definitions.

634 In Fresno, the DWR observation-cell union covers 78.49% of the 495.51-km² rect-
 635 angle but contains 99.71% of its GHSL allocation. Of 598098 allocation units, 1710
 636 fall outside the footprint. At 1 km, uniform allocation gives 9022–9556 outside-
 637 support units, an additional 1.2–1.3 pp of the AOI total. Its large relative multiplier
 638 therefore has a small baseline. Center sampling also varies with grid origin (Supple-
 639 mentary Fig. S19).

640 On the complete Census-block domain, the Census-population-weighted DWR
 641 area-coverage share is 99.19%, versus 99.74% for native GHSL allocation (Sup-
 642 plementary Section S23). This normalized-share contrast reflects relative spatial

644

weights, whereas the 12.20% GHSL–Census total difference is a separate diagnostic. Multiplying all weights by one constant leaves either share unchanged. A separate DWR rate example uses Census-block median observed vertical-rate classes. The annualized six-window 2015–2021 median covers blocks containing 95.70% of complete-block population, of which 64.99% falls below -5 mm yr^{-1} . Different time-window choices change classification; this transfer does not validate Bangkok LOS rates (Supplementary Section S28). Across 96 fixed-region scale/origin/cutoff designs, uniform-minus-native differences are only 0.026–0.186 pp in Los Angeles and 0.051–0.372 pp in Davis–Sacramento. The positive direction transfers, but these small effects do not generalize the approximately 58% primary relative difference (Supplementary Section S29).



3.7 Processing checks and external evidence limits

Full-stack quality rejection leaves 431 of 437 pairs for inversion; the training branch retains 382 of 393 inputs. The full quality mask accepts 97.02% of pixels and 99.71% of the small area’s allocated population. Among pixels with matched-input mean support below 0.2, 83.34% still pass the fixed multi-index mask; the 0.8–1 group has 99.72% acceptance. Low coherence-screened support is therefore not equivalent to final processing rejection (Supplementary Fig. S18). Neither acceptance nor rejection is independent physical truth.

Matched-input support bins also differ in median velocity spread (1.343 versus 1.185 mm yr^{-1}) and inversion residual (0.673 versus 0.368 mm, lowest versus highest bins). These are quality associations, not a binary validity surrogate; legacy-sample confusion counts and period differences are reported in Supplementary Section S17. All 44 held-out edges met the prediction and reference requirements. Pooled held-out LOS displacement root-mean-square error (RMSE) is 0.984 mm across evaluable observations. Training-mask accepted and rejected locations give 0.699 and 5.150

691 mm, respectively; the lowest and highest training-support bins give 2.338 and 0.412
692 mm. On common evaluable observations, time-series differences, training linear
693 velocity and a zero-displacement baseline give 0.984, 11.751 and 11.784 mm. The
694 held-out pairs connect dates already represented in training. Shared acquisition
695 dates and atmospheric phase limit these residuals to internal reconstruction; they
696 do not measure independent velocity accuracy. Exclusions and per-pair results are
697 retained in Supplementary Section S17.

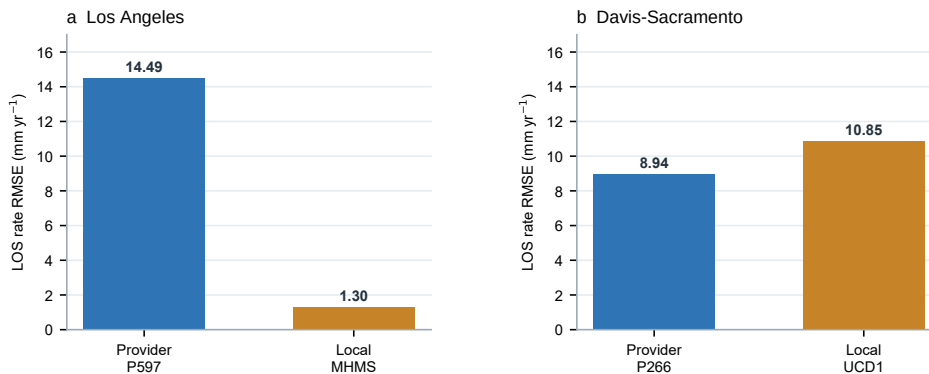
698 On the same processing grid and GHSL field, uniform allocation within 1-km cells
699 increases the deficit by 3.066–3.189 percentage points of AOI population for mean
700 437-pair input support, versus 0.108–0.117 points for the final quality mask. Native
701 deficits are 3.4539% and 0.2944%, respectively. Direct source-cell intersections verify
702 the geometry (Supplementary Section S19). This contrast reflects different support
703 definitions, not an error comparison or observed motion in gaps.

704 Training-only coherence filtering from 0.2 to 0.4 reduces accepted allocation from
705 99.40% to 95.34% and pooled excluded-edge RMSE from 0.491 to 0.388 mm.

706 At 0.7 only 41.04% remains. Other metrics have nonmonotonic responses; all 18
707 fixed-inversion post-filters are reported in Supplementary Section S21, without
708 interpreting residuals as absolute velocity accuracy.

709 External GNSS agreement is mixed. Native-pixel RMSE in Los Angeles is 14.49 mm
710 yr^{-1} under provider reference P597 and 1.30 under location-selected local control
711 MHMS (11 validation stations). Davis–Sacramento gives 8.94 under P266 and 10.85
712 under UCD1 (four stations). Controls are excluded from validation; every sampling
713 radius and both frames are retained. Archived source-version conflicts, temporal-
714 model differences and station dependence limit interpretation. These results cannot
715 be transferred to our Bangkok inversion (Supplementary Section S30).

716 Figure 6 retains both reference-frame choices in each region.



733 Supplementary Fig. S20 and Supplementary Table S29 distinguish account-
734 ing checks, internal reconstruction and external-product comparisons from the
735 unresolved physical validation of the Bangkok inversion.

4 Discussion

4.1 Earth-science informatics implication

The central implication is that public InSAR observability should be treated as part of the information product when deformation maps are translated into exposure estimates. A support mask specifies the observation domain and support available to a downstream analysis. The proposed audit links that support to a traceable population or built-surface allocation quantity.

The matched-input comparisons establish a material change in population-weighted support summaries when fine spatial allocation is replaced by uniform coarse allocation. The effect persists in complete interior cells, so boundary handling does not explain the primary result. Crossing population models and pair selections gives a consistent positive direction across the 16 tested cases, but the relative difference is much smaller with WorldPop than with GHSL. The approximately 58% difference is specific to the primary GHSL case; the crossed comparisons show how the magnitude changes with the population distribution. It does not establish a universal correction factor.

Measurement precision and coherent-scatterer reliability do not fully describe uncertainty in exposure accounting. A product can be accurate where it observes while a population summary remains conditional on that support. Reporting the continuous support deficit makes explicit the mean population allocation unsupported across the specified interferogram pairs. This complements the quality information supplied by the radar processor.

For city analysts, infrastructure planners and hazard-screening researchers, the support mask and population-allocation rule must remain explicit when a public deformation layer is combined with demographic data. Supported and unsupported totals provide a shared interpretation for radar specialists and downstream users who do not revisit the full processing chain.

Reusable products therefore need machine-readable support semantics as well as values. Velocity, coherence, validity and population belong to different data layers. Keeping them separate distinguishes zero exposure from unsupported allocation and allocation outside the domain; the final table should preserve those meanings. Related support-and-allocation questions arise when other Earth-observation products are combined with human-exposure layers or administrative boundaries, including flood, fire and coastal-impact screening. Such applications require their own physical targets and tests; the present experiments establish results only for the stated InSAR and population inputs.

4.2 Relationship to radar and signal-processing practice

The support deficit extends measurement-support and missingness analysis into population-weighted accounting. A processing chain retains coherent or otherwise valid measurements unevenly across space and time; its product therefore need not represent the geographic domain uniformly.

Water, vegetation, surface change and construction make missing support geographically structured. Urban deltas and coastal plains combine dense development, waterways, soft soils and construction zones. Subsidence and exposure intersect in coastal Semarang [53]; time-series radar measures coastal urban subsidence in Jakarta [54]. Wuhan and Karapinar extend the examples to inland urban and basin settings [55, 56]. Where observability correlates with population, the support mask changes the population summary.

The covariance relationship explains the primary overestimation and the opposite errors under density-associated masking. Small aggregate biases under density-independent masks identify conditions in which uniform allocation can be adequate. Preserving finer model structure is useful only insofar as that structure is appropriate to retain; agreement with a native-model reference does not establish household accuracy.

Measurement support and measurement value remain distinct. Low support does not itself indicate subsidence. The primary geographic domain is therefore fixed without a velocity threshold, while the support field records mean area-pair availability. A physical deformation screen would require independently interpretable rate measurements.

Reconstructing deformation in missing locations would require additional observations or assumptions about geology, groundwater, land cover, loading and spatial correlation. Those assumptions may be justified locally but cannot be inferred from a support deficit. Here, numerical correctness is assessed conditionally on specified inputs without asserting deformation in gaps.

Ascending/descending geometries, local GNSS, leveling, piezometry or building observations could support a separate validation or attribution analysis. A verified deformation field would then permit population accounting within physical threshold classes, with unresolved locations still reported. Choosing new measurement sites would additionally require information value and local consequences.

4.3 Use for infrastructure and power-system screening

The workflow could use substations, transmission-line length, distribution corridors or other critical assets in place of E_p . An asset-specific support definition would describe incomplete radar coverage, while risk assessment would also require design, age, redundancy, foundation and maintenance information. The present experiments evaluate neither these assets nor their failure probabilities or power-system risk. Combining support at asset locations or corridors with engineering observations, asset criticality and information value could inform follow-up studies. Vulnerability and intervention priorities require engineering assessment.

4.4 When the audit should and should not be used

The audit is useful when reusing a deformation product with retained support information and allocatable exposure layers, particularly when restricting to visible support changes the denominator. Before local validation, it describes where

the selected population model has weaker support. It cannot determine where measurement or intervention would deliver the greatest benefit. For dense site-level diagnosis, matched leveling, GNSS, borehole, hydrogeological and engineering observations may make public-product support a secondary concern. The audit can document support but cannot replace that evidence. If a product has discarded invalid areas without retaining its mask, omitted support cannot be reconstructed reliably; the appropriate result is a data-limitation statement, not an unsupported-allocation estimate for that domain. A complete-case overlay describes allocation within observed support. Center sampling is inexpensive but can miss subcell structure and boundary coverage. Uniform-area transfer is suitable when its approximation error is acceptable for the intended summary. Building weights add assumptions and may share population-model inputs. Imputation, multiple sensors and ground measurements add deformation information and require separate evaluation; our experiments assess their accompanying support-accounting choices.

4.5 Limitations

The strongest physical limitation is the absence of matched independent GNSS or leveling validation for the Bangkok inversion. A local feasibility audit found that CUSV and CUUT occupy the same InSAR pixel and lack an independently established datum shared with the inversion (Supplementary Section S17). Controlled masks test accounting against conditional references; excluded pair edges test internal reconstruction. California GNSS supplies independent observations for a different product and different regions, with mixed results under two reference-frame choices. None identifies deformation in permanently unsupported Bangkok locations (Supplementary Fig. S20; Supplementary Table S29).

This limitation restricts physical interpretation. A support deficit cannot be converted into confirmed affected population without evidence about deformation at the relevant locations. The controlled experiments instead establish what changes when the same inputs are combined under different allocation rules. The primary Chao Phraya domain inherits the finite source-VLM footprint, so its conclusions do not establish coverage of the whole delta or extend to locations outside that footprint. The additional rectangles likewise do not establish a worldwide magnitude of the effect.

The original majority-observable partition simplifies a continuous support problem. The revised primary analysis therefore retains support fractions and the full population-weighted support distribution. This change addresses threshold information loss, but it does not make coherence-screened pair support equivalent to reliable velocity or to a final processing mask. Sampling variation remains conditional on the archived pair pool; shared acquisitions and atmospheric signals also limit the interpretation of held-out edge residuals. Pair number, seasonal composition and temporal baseline are not isolated experimental factors.

GHSL provides reproducible public inputs but cannot replace local census, cadastral or infrastructure inventories. Population grids may misallocate residents in changing urban areas; built-up products may represent informal settlement, industrial

surfaces and mixed land cover differently. WorldPop, Census, CA-POP and ACS comparisons measure disagreements at stated supports without fully propagating population-product uncertainty or establishing household locations. CA-POP shares Census inputs, and land-cover or built-area weights may share information with GHSL. Incomplete municipal residential classification restricts the common domain, so its results are not population-wide validation. The workflow supports screening, not causal attribution. Subsidence can arise from groundwater withdrawal, sediment compaction, loading, hydrocarbon extraction, tectonics or other processes. National assessments show variation across hydrogeological and urban settings [57]; coastal policy studies show that control benefits and priorities depend on local exposure and management [58]. Attribution requires hydrogeological, geological and geotechnical evidence beyond these inputs. Further physical interpretation also requires independent deformation observations matched to the lead domain and stronger local population or asset data.

Earlier estimates and source attribution.

The submitted 3.772492-million center-majority total is reproducible, but it is not a verified count of people exposed to subsidence in missing areas. The earlier center-majority statistic differs from the mean support deficit defined here; their relationship and input provenance are documented in Supplementary Section S2. We have removed the exploratory Iran branch from quantitative applications. Its archived rate and mask correspond to the 2014–2020 product associated with Haghshenas Haghighi and Motagh (2024). No Iran result or new/old Iranian product comparison contributes to the quantitative conclusions. Supplementary Section S3 documents the attribution correction and distinguishes the newer products of Payne et al. [12, 59, 60].

5 Conclusions

Spatial aggregation changes population-weighted InSAR support summaries by omitting the within-cell population–support cross moment. Fixed-input and complete-interior comparisons establish this effect. Its direction persists across the tested population models and pair selections, while its magnitude depends on the inputs and study area. Controlled missingness identifies conditions under which the error reverses, aggregate bias is small or substantial allocation remains unestimated. The published-support and regional comparisons show that transfer also depends on the support definition; external GNSS agreement differs between regions. Population-weighted reporting should therefore specify its domain and support field, preserve extensive quantities and their cross moments, and report sensitivity to allocation and sampling. Deformation in unsupported locations remains unknown.

Screening with this workflow requires a declared domain, population model, support field and allocation rule; summaries restricted to observed support must be labelled accordingly. Site diagnosis and intervention still require denser geodetic, demographic and geotechnical evidence.

Data Availability

Public input data are obtained from COMET LiCSAR, GHSL, WorldPop, ESA WorldCover, the U.S. Census Bureau, California DWR, the archived ARIA California products, NGL and City of Fresno municipal GIS. Exact source-product identifiers, acquisition methods and input checksums are documented in the Supplementary Information and accompanying reproducibility materials. Source-provider terms govern access to and reuse of third-party inputs. Frozen processed analysis inputs, full reference and replicate tables, figure source data and the manuscript number ledger are publicly available with the versioned scientific reproduction package.

Code Availability

The scientific reproduction package is publicly available at the fixed GitHub release [v0.6.2](#). It contains the experiment code, configurations, input manifests and checksums, processed analysis inputs, full reference outputs, environment records and reviewer reproduction instructions. Companion assets supply portable inputs and selected complete source products for S15, S20–S21, S27, S29–S30, with file-level checksums. The complete ACS files and official Census source archives used in S25 remain available at v0.4.1. The reviewer guide provides reproduction and numerical-comparison commands for the processed inputs and reference outputs retained from release v0.6.0. The current manuscripts, two portable local audits and a section-to-output index are identified separately under `revision_2026/submission_20261003/`; these current documents supersede earlier manuscript snapshots in the archive. Other provider-controlled original products and the complete interferogram processing stacks are obtained separately through the documented acquisition routes. The earlier DOI [10.5281/zenodo.21444768](#) archives v0.3.0; a revised DOI deposit remains pending.

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1163 Additional Information

1164 Competing interests

1165 The authors declare no competing interests.

1166 Figure legends

1167 **Figure 1.** Workflow for population-weighted support accounting. The comparison fixes the
1168 geographic domain, support field and allocated total. It compares uniform coarse allocation
1169 with preservation of the native population–support cross moment. Neither branch infers
1170 motion in unsupported locations.

1171 **Figure 2.** Chao Phraya population and built-surface support deficits in the fixed geo-
1172 graphic domain at $q = 0.3$. Red bars show uniform allocation within 1-km cells averaged
1173 across four origins; narrow whiskers span those origins. Blue bars show the native-model
1174 reference. These are conditional model-allocation quantities, not unique residents or con-
1175 firmed subsidence exposure. The submitted 3.77-million statistic uses a different domain
1176 and binary selection; its lineage is preserved in Supplementary Section S2.

1177 **Figure 3.** Continuous support sensitivity. Curves give the fraction of allocated pop-
1178 ulation or built-surface quantity with native-pixel mean support below t , for joint
1179 unwrapped/coherence cutoffs $q = 0.2, 0.3, 0.4$. The domain and allocated totals are iden-
1180 tical across cutoffs. The display uses 101 threshold values; the exact empirical-distribution
1181 integral is checked separately. No velocity threshold determines these curves.

1182 **Figure 4.** Pair-sampling sensitivity. (a) Means and central 95% ranges from 100 draws at
1183 each size within the fixed 224-pair pool, at $q = 0.3$. At 224 pairs the whole pool is selected
1184 and the range collapses. (b) Population and built-surface deficit shares in the fixed nested
1185 selections. The nested 28 pairs differ from the legacy 28-pair sample. Ranges condition on
1186 the archived pool; neither panel supplies calibrated uncertainty for household population or
1187 true deformation.

1188 **Figure 5.** Regional transfer under specified support fields. (a) Uniform-minus-native popu-
1189 lation support deficits for 1-km cells, in percentage points of each fixed AOI total; whiskers
1190 span four grid origins. (b) Native population-weighted support in the same domains. Chao
1191 Phraya uses sampled mean area–pair support at $q = 0.3$; Fresno uses the nonduplicated
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DWR observation-polygon union; Los Angeles and Davis–Sacramento use their documented ARIA quality-support definitions at the displayed 0.3 cutoff. The support objects differ, and the effects are not pooled or interpreted as a universal correction factor.

Figure 6. External GNSS consistency checks. Native-pixel California InSAR–GNSS LOS-rate RMSE is shown for the archived provider frame and the location-selected local frame. Control sites are excluded from the 11 Los Angeles and four Davis–Sacramento validation stations. Both reference choices are retained. Temporal-model, source-version and station-dependence limitations are detailed in Supplementary Section S30. These comparisons do not validate the separate Bangkok inversion or genuinely missing deformation.

Table legends

Table 1. Native-model support deficits in the primary geographic domain and legacy 28-pair sample. Each percentage uses its stated population or area denominator.

Table 2. Checks on support accounting, grouped by the alternative explanation or information limitation they address. The table summarizes experiments detailed in the main Results and Supplementary Sections S4–S30; it is not a combined validation score.